

Ali Nawaf

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Education

Case Western Reserve University | Cleveland, OH

MS in Machine Learning and Artificial Intelligence.

May 2027

BS in Computer Science, Secondary Major: Mathematics.

May 2027

Coursework: AI, ML, Networks, Security, HPC, Stochastic Calculus, Probability, Algorithms, Linear Algebra

Deans Honor list, Davis Scholar (United World Colleges), Scholar Of Distinction award

Skills

Languages: C, Python, Java, SQL, Swift, JavaScript (Node.js, Next.js), TypeScript.

Machine Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Keras, Retrieval Augmented Generation (RAG).

Cloud & DevOps: Azure, AWS, Power BI, SLURM, HPC, Kubernetes, AWS, Lambda functions, Linux, REST APIs

Experience

AI & ML Engineering Intern | Trek Health | San Ramon, CA

Dec 2025 – Present

- Automated autonomous classification and content retrieval, across a production system of 7 million data chunks for thousands of active users, by architecting a production-grade RAG and candidate re-ranking pipeline.
- Boosted system recall and content discovery quality, as measured by a 105% increase, by engineering an agentic framework featuring multi-hop interfile discovery, query transformation, candidate scoring, and corrective rag.

ML Research Assistant | Houston Methodist Hospital | Houston, TX

Aug 2025 - Present

- Independently researched and developed a novel hypothesis for cardiovascular risk prediction, applying a pioneer ML model to test the theory against a real-world CT scan dataset, culminating in **90%** predictive accuracy
- Engineered a signal processing pipeline to digitize raw ECG images, implementing **open-source** models and post processing algorithms to extract meaningful signals and enhance fidelity for downstream ML analysis

AI SWE Intern | Heads-up Hockey | Waterford, VA

Dec 2024 – Aug 2025

- Automated computer vision annotation pipeline, reducing time by **70%** using **SAM2** and a **YOLO model**.
- Built an interactive **Swift UI** educational game called “**Heads up Crush**” with on-device **AI inference**. Which increased userbase by **100%** upon release
- Collaborated with a team of 3 engineers to develop the company’s pioneer Hockey stick-handling **AI model** which improved accuracy in unseen testing by **30%**
- Engineered an **end-to-end AI agent** by implementing data **pre-processing** and **post-processing** to extract goal counts, speed, and trajectory, achieving **70%** accuracy in shot trajectory prediction using the iphone’s camera

AI SWE Intern | Eaton | Cleveland, OH

Jan 2025 – May 2025

- Built a fully automated pipeline processing of **10k Salesforce** records per month hosted completely on **Azure**
- Implemented batch analytics workflows, cutting latency by **35%** and lowering operational costs
- Delivered business insights using **Power BI**, built a complete **Agentic RAG system** for information intelligence

Projects

Dispatch: Unified AI Agent Orchestrator | Next.js, FastAPI, Supabase, Tauri, Python, TypeScript, Twilio |

Enabled **hands-free remote programming** allowing users to "call" their computer to execute code converting voice intents into local terminal commands and agentic actions

AI Medical Labeler | PyTorch, Python, API, git, GitHub, Docker, Networking | [link](#)

Reduced manual image annotation time by **60%** by developing a Python-based tool that integrated with the Label-Studio API to pre-label images using **SAM2** model

Variational Autoencoder (VAE) Implementation | Python, PyTorch, TensorFlow, NumPy, MNIST, Matplotlib | [link](#)

Designed and implemented a **generative deep learning model** to reconstruct images and generate new synthetic data